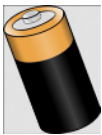


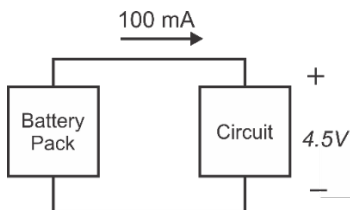
1. Answer the following.
 - a. Find the charge, in C (Coulombs), of 5.31×10^{20} electrons.
 - b. Find the current from a steady movement of the electrons found in (a) over 4 seconds. The electrons are moving from left-to-right (your current direction can be "left-to-right" or "right-to-left")
2. Convert the following to engineering notation using common prefixes with factors of 3 in the exponent. (For example, $2200 \times 10^{-4} \text{ A} = 220 \times 10^{-3} = \underline{220 \text{ mA}}$)
 - a. 0.045 W
 - b. $2000 \times 10^{-5} \text{ J}$
 - c. 0.1 secs.
 - d. 18,000 V
 - e. 0.00031 A
3. An electric car is equipped with a 175 hp motor. Answer the following questions using the Basic Conversion factors on the reference sheet
 - a. How many kW are required to run the motor if we assume 100 percent efficiency in converting electric power to mechanical power?
 - b. If the efficiency of the motor is only 80 % (0.80), how many kW are now required to run the motor?
 - c. For 3-a, how much energy (in J) is expended if the motor is run continuously for 3 hours?
 - d. If a single lead-acid battery has a 430-kilowatt-hour storage capacity, how many batteries are required for 3-c?
4. How many joules does a 60 W (watt) incandescent light bulb consume in 1 hour?
5. Search the Digikey.com website and find the datasheet for a non-rechargeable AA-type alkaline battery which includes a Capacity rating (in amp-hour). How many hours will a single AA battery be able to power a device that draws 1mA of current? Include the datasheet as part of your homework submission.
6. Answer the following questions regarding battery requirements in a circuit.



A single non-rechargeable D-cell battery has the following specifications:

- 1.5 V
- Capacity = 19 A-hr

A circuit requires 4.5 V and 100 mA.



- A) What is the minimum number of batteries required for this circuit to operate continuously for three weeks? (batteries can be connected in series and/or parallel).
- B) Draw a schematic diagram for your battery pack in (A).